PURDUE
UNIVERSITY
Department of Physics & Astronomy

PHYS 272
Electric & Magnetic Interactions
Spring 2026
Course Website

Instructor & Coordinator: Prof. John Peterson

Instructor: Lect. Laura Urba

Instructor: Prof. Qing (Shilo) Xia

Instructor (Indianapolis): Prof. Jeff Gerber

Coordinator of Digital Instruction: David Huckleberry

Textbook and Required Materials:

Matter & Interaction II – Electric and Magnetic Interactions, 4th Edition, Ruth Chabay and Bruce Sherwood, Publisher: Wiley, or any of the options listed on the course website.

Physics 27200 Laboratory Manual will be posted each week on the Brightspace website.

iClicker Cloud (Student App) Use your phone or laptop. DO NOT buy an iClicker device! (Check Brightspace under Content for more information)

Scientific calculator without internet capabilities for exams

Course Webpage:

You will find everything you need on the Brightspace website and course website, including this syllabus, textbook information, course announcements, Help Center hours, and other important resources.

Course Components:

Lecture

Get points for coming to lecture! Expect iClicker questions every lecture which count toward Lecture Participation. You can get 10 points per lecture: 8 for answering half of the polls (attendance component) and up to 2 points for correct answers (performance component). You will be given sufficient time to answer the questions, and it provides feedback during class.

Find information on how to set-up your iClicker Cloud account for the course on Brightspace under Content. DO THESE ASAP! We will practice iClicker questions during the first week, and then consider the top 21 scores of the final 25 lectures for the final grade. Boilercast recordings of the lectures will be available on Brightspace under Course Tools/Kaltura Media Gallery.

Reading

Make your life easier! Read the assigned parts of the text *before* each lecture. Lecture slides will be posted before each class. When you see the "?" in the text, stop and answer the question before continuing. Both reading the material and coming to lecture significantly will improve your performance.

Homework

Homework will be assigned on Brightspace using a system called WileyPlus. The system contains a compendium of problems that we will use for our course. Two homework assignments will be due each week with DEADLINE: 11:59 PM on the day indicated in the course schedule.
LATE POLICY: 50% credit is given until 24x7=168 hours after the deadline and within the allowed number of attempts. No credit will be given after that! You will have 10 attempts per question in each homework assignment. Homeworks 1 to 24 are "must do" homework assignments and we will consider your top 20 scores. Homework 25 is for extra credit.

Recitation

Weekly recitation sessions meet for one hour. You are required to attend the recitation section assigned to you by the Registrar**. Recitations begin the *first week* of class.** After the first week there are 12 recitations and we will consider your top 10 scores for the final grades. Students that (without a valid excuse) arrive more than 15 minutes late to a recitation will receive a zero grade. Students not actively working or disrupting during recitations will have a two point deduction on that recitation grade.

Laboratory

Weekly laboratory sessions meet for two hours. You are required to attend the lab section assigned to you by the Registrar**. Labs begin the *first week* of class.** After the first week there are 12 labs and we will consider your top 10 scores for the final grades. Students that (without a valid excuse)

arrive more than 15 minutes late to a lab will receive a zero grade. Students not actively working or disrupting during labs will have a two point deduction on that lab grade.

Exams

All exams are in-person and closed-book. There are three exams, two midterms and a final. All three exams are required (see makeup section). The final Exam is cumulative. Students need to bring their Purdue ID, a pencil, and a scientific calculator without internet capabilities to all exams. Cheat Sheets are NOT allowed. An equation sheet will be provided with each exam. The two midterm exams are scheduled to be given on:

Exam 1: Monday, February 16th, at 8-9:30 pm in Elliott Hall
Exam 2: Monday, April 6tt, at 8-9:30 pm in CL50 224, Lily 1105, Walc 1055, Phys 112

Indianapolis campus students exams will be held at the same time in NUC 103 & 108

The final exam schedule can be found in the following link:
https://roomschedule.mypurdue.purdue.edu/Timetabling/exams.do

The date for the final exam schedule is set by the Registrar later in the semester. Do not plan to leave campus before this date!

Grades:

Grades will be determined from points earned in each of these categories:

Midterm Exams (2)	30%
Final Exam	20%
Labs	20%
Homework (WileyPlus access via Brightspace)	15%
Recitation	10%
iClicker (Lecture Participation)	5%
TOTAL	100%

We will use an absolute scale to set the grades as given in the table below. *At the end of the semester, if the grades are lower than we think is fair then we may lower the grade thresholds*, but we will not raise the thresholds. (This means if you get at least 89%, you are guaranteed to get an A- or better.)

Grade	Percentage
A+	95%
A	92%
A-	89%
B+	85%
B	82%
B-	79%

C+	75%
C	72%
C-	69%
D+	65%
D	62%
D-	59%
F	<59%

Please check your scores at any time. It is your responsibility to ensure that the grades are accurate. *Report any errors to your recitation instructor and the professor within 2 weeks from when the grades are recorded.*

iClicker Scoring: Points can be earned for every lecture (except for the first week). Please report technical issues to innovativelearningteam@purdue.edu

Learning Outcomes:

By the end of the course, you will be able to:

- Understand the concepts of electric field, magnetic field, and an atomic description of matter to describe polarization.
- Calculate electric fields and electric potentials produced by charge distributions using calculus-based mathematical tools.
- Analyze and solve electrical circuits.
- Calculate magnetic fields, magnetic forces, induction, and related topics using calculus-based mathematical tools.
- Have a general understanding of the Maxwell's equations and the electromagnetic radiation.
- Construct computational models using VPython and perform experiments based on the physics principles learned in the lectures.

Use of artificial intelligence (AI) or Large Language Models (LLM):

AI and LLM are **not allowed** to use in this course.

Help:

This course is challenging. Plan to study hard! Here are some ways to get a better grade:

- Go to the Physics Help Center, Room 290 in the physics building. It is staffed during regular hours all day long on Tuesday and Friday by Teaching Assistants who are trained in this material.
- Ask your instructor during lab, recitation, or lecture.
- See the instructor or TA's during office hours. We are happy to work through physics problems with you.
- Use the buddy system: study with a friend in class. You are encouraged to work on homework together – discussing ideas and concepts reinforces the material for everyone

involved in the conversation. Just be sure that what you turn in is your own work that you fully understand.

- For Indianapolis Campus Students: This course is supported by the Academic Success Center (ASC) Peer-Led Tutoring Program. You can schedule sessions through BoilerConnect. Sessions must be scheduled 24 hours in advance. This free service is a great way to practice the material, reinforce challenging material, and review key concepts. Learn more & schedule a session here: ASC Tutoring. The ASC provides non-content specific support through Study Skill Consultations & Peer Success Coaching. Learn more at ASC Indianapolis!

Absences and Exempt (EX) Grades:

For the clicker questions, recitations, labs, and homeworks, you can potentially receive a exempt grade. To receive an exempt grade, we require either: 1) a note from ODOS for serious illness or family emergency (see next paragraph), or 2) some documentation for a school sponsored activity that conflicts with the event (sports travel, clubs, conference, etc.).

For a serious illness or a family emergency, you may request an exempt grade so missing that item will not count against your final grade. To do this, **you should contact ODOS directly**, and send them documentation of the illness or grief (**Do not send this documentation to the TAs or Instructors**). Then ODOS will send us an email after they verify the illness or death. Please contact ODOS as soon as is reasonable, so we receive the ODOS note within 3 weeks of the event.

In many cases of minor illness, ODOS will not be able to issue a MEAPS notice. So for minor illnesses, we have made the policy to take the top scores of the clicker (21 out of 25), homework (20 out of 24), labs (10 out of 12), and recitation (10 out of 12). Therefore, for minor illness you do not have to contact anyone or have any documentation. If you have more absences than the policy allows, you should be able to obtain a notice from ODOS.

If you do obtain a MEAPS notice from ODOS, you should follow up promptly with your TA or instructor to let them know which graded items you will request an excused grade. ODOS will send us an email verifying the illness or death, but it is not always clear which grades you missed.

Missing EXAMS and Makeup EXAMS:

Since the exams are a critical part of the grade, we will give a makeup exam to you if you have a valid excuse for missing the exam. Make-up exams can be requested for the following reasons: documented schedule conflicts, religious observances, documented attendance to a Purdue sponsored event, grief verified by ODOS, illness verified by ODOS. It is very important that you email your instructor and the coordinator as ***soon as possible*** (if possible, in advance of the exam). If you are unable to obtain a ODOS note or other documentation, we will consider a makeup exam depending on the situation and you should send as much information and/or documentation as possible (again it is important to let us know as soon as possible, if you are unable to be there).

DRC Accommodations:

If you receive notification from the Dean of Students that you are entitled to accommodation, supported by a DRC Letter, please inform your instructor(s) as soon as possible. They will then confirm that they have received notification from DRC regarding your accommodation and if needed, schedule a meeting to determine proper actions.

If your accommodations include **EXAM TAKING**, please note the following responsibilities to receive those properly:

- Must schedule a testing session with Purdue Testing Services (PTS) for each exam listed on the course schedule. NOTE: PTS requires 7 days lead time for scheduling. They will deny any request made within 7 days of the scheduled exam. It is highly recommended that on the day you get approved for exam accommodation, you immediately schedule the exam testing sessions you need, as they have limited availability.
- Course personnel will then provide the Exam (in proper format) to PTS ahead of your scheduled test session and coordinate scoring.
- If you do not properly secure an accommodated testing session ahead of the scheduled exam day, you may still take the exam without accommodation with the rest of the class
- If you do not show up for your accommodated exam, you will not be allowed a makeup exam unless for a valid documented reason.

Academic Honesty:

You are encouraged to work on homework together – discussing ideas and concepts reinforces the material for everyone involved in the conversation. Just be sure that what you turn in is your own work that you fully understand.

The following are examples of cheating:

- Any effort to present someone else's work as your own, or allowing your work to be presented as someone else's.
- Having someone else solve problems assigned for you.
- Entering iClicker responses for someone else.
- Have someone sign attendance for you or sign attendance for an absent student in a Lab or Recitation.
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If a student is found cheating, the incident will be thoroughly investigated by the Dean of Students, and appropriate disciplinary actions will be taken. In serious cases the Dean may suspend or expel the student from the university.

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breeches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information that is submitted provides the greatest opportunity for the university to investigate the concern.

Possible Changes:

In the event of a **major campus emergency**, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances. You can get information about changes in this course at the Brightspace website.

Emergency Response:

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a daily basis.

- To report an emergency, call 911.
- To obtain updates regarding an ongoing emergency, sign up for Purdue Alert text messages, view www.purdue.edu/ea.
- There are nearly 300 Emergency Telephone Systems throughout campus that connect directly to the Purdue University Police Department (PUPD). If you feel threatened or need help, push the button and you will be connected to the PUPD.
- If we hear a fire alarm during class we will immediately suspend class, evacuate the building, and proceed outdoors. Do not use the elevator.
- If we are notified during class of a Shelter in Place requirement for a tornado warning, we will suspend class and shelter in [the basement].
- If we are notified during class of a Shelter in Place requirement for a hazardous materials release, or a civil disturbance, including a shooting or other use of weapons, we will suspend class and shelter in the classroom, shutting the door and turning off the lights.

Please review the Emergency Preparedness website for additional information. http://www.purdue.edu/ehps/emergency_preparedness/

Accessibility:

Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let your instructor know so that you can discuss options.

You are also encouraged to contact the Disability Resource Center at: *drc@purdue.edu* or by phone: 765-494-1247. (The weblink to the DRC is also available below.)

For accessibility concerns beyond this, the **Office of Institutional Equity** is responsible for ensuring Americans with Disability Act compliance, can be contacted with any accessibility concerns at:

Phone: (765) 494-7253
Email: equity@purdue.edu
TTY: (765) 496-1343
Website: http://www.purdue.edu/ethics/oie/

Counseling and Psychological Services:

- If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try WellTrack. Sign in and find information and tools at your fingertips, available to you at any time.
- If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 am- 5 pm.
- If you find yourself struggling to find a healthy balance between academics, social life, stress, etc. sign up for free one-on-one virtual or in-person sessions with a Purdue Wellness Coach at RecWell.Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is completely free and can be done on BoilerConnect. If you have any questions, please contact Purdue Wellness at evans240@purdue.edu
- If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS office of the second floor of the Purdue University Student Health Center (PUSH) during business hours."

Basic Needs Security:

Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students for support. There is no appointment needed and Student Support Services is available to serve students 8 a.m.-5 p.m. Monday through Friday. Considering the significant disruptions caused by the current global crisis as it related to COVID-19, students may submit requests for emergency assistance from the Critical Needs Fund.

Nondiscrimination Statement:

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. A hyperlink to Purdue's full Nondiscrimination Policy Statement is included in our course Brightspace under University Policies.

CAPS Information:

Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at (765)494-6995 and http://www.purdue.edu/caps/ during and after hours, on weekends

and holidays, or through its counselors physically located in the Purdue University Student Health Center (PUSH) during business hours.

Diversity and Inclusion:

Purdue's Nondiscrimination Policy Statement:
https://www.purdue.edu/purdue/ea_eou_statement.php

Disability Support and Religious Observance:

Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, please let us know as soon as possible so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247. If you anticipate missing exams / assignments due to participation in religious holidays or days of religious, ethnic, or civic observances, please notify us in advance to discuss alternative arrangements.

Title IX Resources:

The purpose of Title IX is to help foster safe and respectful university environments that better protect students, faculty and staff from incidents of sex-based discrimination and sexual harassment, including sexual violence, relationship violence and stalking. Resources for West Lafayette campus: https://www.purdue.edu/sexual_assault/titleix/overview.php

Schedule:

Week	Dates	Chapters	Lectures	Lab	Rec
1	Jan 12-16	13.1-13.6	1 & 2	0 (Ungraded)	0 (Ungraded)
2	Jan 19-23	14.1-14.3	3 on Wed (None on Mon)	1	1
3	Jan 26-30	14.4-14.6, 15.1-15.3	4 & 5	2	2
4	Feb 2-6	15.4-15.11	6 & 7	3	3
5	Feb 9-13	16.1-16.7	8 & 9	4	4
6	Feb 16-20	16.8-16.11	10 on Wed (None on Mon)	None	None
7	Feb 23-27	17.1-17.7	11 & 12	5	5
8	Mar 2-6	17.8-17.11, 18.1-18.4	13 & 14	6	6
9	Mar 9-13	18.5-18.10, 19.1-19.3	15 & 16	7	7
10 (Spring Break)	Mar 16-20	None	None	None	None
11	Mar 23-27	19.4-19.11	17 & 18	8	8
12	Mar 30-Apr 3	20.1-20.6	19 & 20	9	9
13	Apr 6-10	20.7-20.11	21 on Wed (None on Mon)	10	10
14	Apr 13-17	21.1-21.7	22 & 23	11	11
15	Apr 20-24	22.1-22.7	24 & 25	12	12
16	Apr 27-May 1	23.1-23.7	26 & 27	None	None
Finals	May 4-8				

Week	Dates	Homework	Exam
1	Jan 12-16	None	
2	Jan 19-23	1&2 Due Fri at 11:59 PM	
3	Jan 26-30	3&4 Due Fri at 11:59 PM	
4	Feb 2-6	5&6 Due Fri at 11:59 PM	
5	Feb 9-13	7&8 Due Fri at 11:59 PM	
6	Feb 16-20	9 Due Fri at 11:59 PM	Midterm #1 Mon 2/16 Elliott, 8-9:30 pm PIN: NUC 103&108
7	Feb 23-27	10 Due Fri at 11:59 PM	
8	Mar 2-6	11&12 Due Fri at 11:59 PM	
9	Mar 9-13	13&14 Due Fri at 11:59 PM	
10 (Spring Break)	Mar 16-20	None	
11	Mar 23-27	15&16 Due Fri at 11:59 PM	
12	Mar 30-Apr 3	17&18 Due Fri at 11:59 PM	
13	Apr 6-10	19&20 Due Fri at 11:59 PM	Midterm #2 Mon 4/6 8-9:30 pm (CL50 224,Lily 1105,Walc 1055,Phys 112) PIN: NUC 103&108
14	Apr 13-17	21&22 Due Fri at 11:59 PM	
15	Apr 20-24	23&24 Due Fri at 11:59 PM	
16	Apr 27-May 1	25 Due Fri at 11:59 PM (Extra Credit)	
Finals	May 4-8		Final (TBD)